Lab – 1: Architecture Theory Assignment Answers

Aim: Understanding the Evolution and comparison of different architectures.

->Assignment :

1. What is monolithic architecture? Mention the advantages and disadvantages.

Monolithic architecture is a traditional software design pattern where all components—such as user interface, business logic, and data management—are packaged and deployed as a single tightly integrated unit.

Advantages:

• Simple to develop initially since everything is in one place.

• Easy to deploy because only one unit is involved.

• Fast internal communication as all modules run in a shared memory space.

Disadvantages:

• Difficult to scale individual parts of the system.

• Hard to maintain due to tight coupling between components.

• Any small change requires rebuilding and redeploying the entire system.

• Poor fault isolation—one failure can impact the whole application.

2. What is client server architecture? Mention the advantages and disadvantages.

Client–server architecture separates the system into a client that requests services and a server that provides them.

Advantages:

• Centralized security at the server.

• Reduced client maintenance; updates occur on the server.

• More powerful processing at the server side.

Disadvantages:

• Tight coupling between client and server.

• Heavier processing load on clients in some models.

• Mostly synchronous and stateful, leading to dependencies.

• Server updates may require client adjustments.

3. What is the difference between single-tier and two-tier client server architectures? Mention the applications of both.

Single-tier Client–Server:

• The client is a thin or dumb client with minimal processing.

• The server handles most or all of the application logic and data.

• Supports both synchronous and asynchronous communication.

Two-tier Client–Server:

• The client is a fat or intelligent client with a significant portion of business logic.

• The server mainly handles database-related operations and some rules.

• Communication is typically synchronous.

Applications:

• Single-tier: Web browsing, thin-client environments.

• Two-tier: Enterprise database applications, desktop-based DB systems.

4. What is multi-tier client server architecture? Mention the advantages and applications.

A multi-tier architecture separates software into three or more layers—presentation, business, and data—each running independently.

Advantages:

• Better scalability and maintainability due to separation of concerns.

• Business logic can be reused across multiple applications.

• Reduced load on client systems.

Applications:

• Large enterprise systems.

• E-commerce platforms.

• ERP and banking systems.

5. What is distributed internet architecture? Compare and contrast with other architectures and identify the applications.

Distributed Internet Architecture extends multi-tier design to web-based systems. It uses web browsers as clients, web servers for processing logic, and database servers for data storage.

Comparison:

• Uses browser-based thin clients.

• Employs internet technologies like HTML and HTTP.

• More loosely coupled than traditional client–server.

Applications:

• Web portals.

• E-commerce websites.

• Large-scale distributed applications.

6. What is hybrid web-services architecture? Identify the difference with other architectures and mention the applications.

Hybrid architectures incorporate web services to wrap existing components, without following full SOA principles.

Differences from SOA:

• Uses point-to-point connections instead of loosely coupled messaging.

• Does not follow full service-orientation principles.

• Limited interoperability.

Applications:

• Integrating legacy systems.

• Temporary transition setups toward SOA.

7. What is microservice architecture? Identify the need and applications.

Microservice architecture breaks an application into small, independent services that communicate through lightweight APIs.

Need:

• Independent deployment of components.

• Increased scalability and flexibility.

• Faster development cycles.

Applications:

• E-commerce services.

• Large SaaS platforms.

• Streaming and OTT services.

8. Identify and list the differences between Client-Server architecture and Service Oriented Architecture.

Client–Server:

• Tight coupling.

• Centralized logic often placed in client or server.

• Synchronous and stateful communication.

SOA:

• Loosely coupled services.

• Logic distributed across autonomous services.

• Supports synchronous and asynchronous messaging.

• Uses standardized technologies like XML and SOAP.

9. Identify and list the differences between Distributed Internet architecture and Service Oriented Architecture.

Distributed Internet Architecture:

• Components use proprietary APIs.

• Mostly synchronous communication.

• HTML and HTTP are common but optional.

SOA:

• Standardized service interfaces.

• Encourages asynchronous and stateless communication.

• Built strictly on XML and web service standards.

SOA provides greater interoperability than DIA.

10. Identify and list the differences between Hybrid web-services architecture and Service Oriented Architecture.

Hybrid Architecture:

• Uses web services as wrappers.

• Relies on point-to-point connections.

• Not fully service-oriented.

SOA:

• Supports different messaging patterns.

• Fully follows principles of service orientation.

• Offers high interoperability and loose coupling.

11. Identify and list the primary differences between SOAP based and RESTful web services.

SOAP:

- A protocol with strict standards.
- Uses XML exclusively.
- Heavyweight but very secure with WS-Security.
- Supports stateful or stateless exchanges.

REST:

- An architectural style.
- Allows many data formats (JSON, XML, etc.).
- Lightweight and fast.
- Stateless and ideal for web and mobile APIs.